Background Information Framework for an Agentic AI Actuarial Project

Implementation Guide: Cyber Data Breach Frequency & Severity Pricing Engine

Timeline: 3-Week Project Cycle **Domain:** Capital & Risk / Commercial P&C **Architecture:** ReAct Agent Framework

1. Executive Summary & Project Intent

This comprehensive background guide provides the mathematical, actuarial, and computer science foundations required to construct an autonomous, end-to-end **Agentic AI Cyber Insurance Pricing Engine**. Given a strict three-week delivery window, this project steers completely clear of messy, unstructured data structures—such as corporate policy PDFs or narrative incident logs. Instead, it relies strictly on structured, tabular historical records mapping corporate data breaches by industry sector, financial scale, and record exposure.

The finalized AI system operates as an autonomous risk consultant. It accepts natural language queries containing a company's descriptive profile (e.g., sector vertical and gross revenue), programmatically explores a localized database utilizing dynamic code generation, executes classical loss-cost frequency and severity equations, and returns a mathematically robust, actuarially sound insurance premium.

2. Actuarial Framework: Modeling Cyber Risk

Cyber insurance pricing diverges sharply from traditional physical perils due to its systemic nature and the lack of historical stability. However, the foundational method for pricing a basic data breach notification policy relies upon the classical collective risk model framework [1]. The pure premium represents the expected value of total losses over a specified policy period.

2.1 Frequency and Severity Decomposition

Total aggregate losses are factored into two separate, independent components: annual claim frequency and claim severity. Let N be a random variable representing the number of data breach incidents a firm experiences per year, and let X be a random variable representing the severity (financial loss or record count) per incident. The aggregate loss S is modeled as:

$$S = \Sigma_{i=1}^{N} X_i$$

Under standard actuarial assumptions where claim frequency N and individual severities X_i are mutually independent, the expected pure premium $E[S]$ simplifies to the product of expected frequency and expected severity:

$$E[S] = E[N] \times E[X] = \lambda \times \mu$$

Where λ represents the localized annual breach probability for a given industry sector, and μ represents the expected mean cost (or record count, denoted as $PwnCount$) per breach event.

2.2 Risk Loading and Expense Strategy

To transition from a theoretical pure premium to an actuarially sound commercial rate, a safety loading factor must be systematically appended to account for process volatility, parameter uncertainty, and administrative overhead [2]. The gross written premium (GWP) is mathematically formalised as:

$$GWP = E[S] \times (1 + \theta) + \varepsilon$$

Where θ represents the risk load percentage (typically scaled between 15% and 30% for highly volatile lines like cyber risk to protect against catastrophe accumulation) and ε handles static operational or transaction overhead.

3. Tabular Data Foundations & Processing

To ensure high execution fidelity within the compressed timeline, the model leverages structured relational data tables where industry sectors serve as the foreign keys.

3.1 Primary Data Elements

The underlying tabular data contains several highly clean categorical and continuous variables that allow the agent to segment exposures precisely:

- **Industry Sector:** Categorical divisions (e.g., Healthcare, Financial Services, Retail, Technology). This is the primary covariate driving baseline risk variations [3].
- **Company Scale / Revenue Tier:** Continuous numerical variables used by the agent to normalize loss sizing across different firm magnitudes.
- **Records Leaked ($PwnCount$):** The primary volume measure of a breach, detailing the exact number of compromised customer rows.
- **Financial Indemnity / Payout:** The historical net cost incurred by the insurer for breach notifications, regulatory compliance, and legal defense.

3.2 Recommended Datasets

Dataset Name	Repository Source	Actuarial Utility
Cybersecurity Breach Impact Dataset	Kaggle Repository	Provides precise mapping of real-world historical breach events to corporate revenue, industry types, and explicit dollar payouts.
Cyber Attacks Dataset - UMD	University of Maryland / GitHub	Excellent time-series log of infrastructure attack vectors and records compromised, optimal for empirical frequency derivations.

4. Core Software & Agent Architecture

The shift from a static LLM prompt wrapper to an autonomous risk assistant is achieved through a **ReAct (Reasoning and Acting) loop** framework [4]. The agent does not calculate premiums inside its weights; instead, it writes and executes Python code to interrogate the tabular data deterministically.

> **The Three-Step ReAct Archetype**
>
> 1. **Reasoning:** The LLM parses the natural language user profile, mapping text tokens to known statistical categories (e.g., mapping "hospital system" to the "Healthcare" column value).
> 2. **Acting:** The agent translates its reasoning into a precise `pandas` filtering execution block.
> 3. **Observing:** The runtime environment runs the code, passing statistical outputs back into the LLM context for premium rendering.

4.1 Systemic Framework Options

For a rapid three-week turnaround, you should avoid building custom multi-agent network state machines from scratch. Instead, adopt one of these highly streamlined, production-ready open-source architectures:

- **LangGraph:** Best option for enforcing strict, non-cyclical state loops. It allows you to build a state graph ensuring that the agent *must* execute a validation script before returning a final rate.
- **CrewAI** / **AutoGen:** Ideal if you wish to separate tasks by function (e.g., assigning one agent to act as the "Data Retrieval Query Expert" and a second to act as the "Actuarial Pricing Officer").

5. Implementation Roadmap (3-Week Execution Plan)

To guarantee full completion before the deadline, you must aggressively separate the data layer, the execution tools, and the conversational agent interface.

Week 1: Data Engineering & Tabular Sanitation

Download your chosen structured CSV files. Write a baseline Python module (independent of any AI framework) that pre-aggregates mean breach counts and frequencies by industry sector. Handle missing values explicitly, drop corrupted records, and ensure your `pandas` indices are clean. Set up your local execution directory with a structured schema.

Week 2: Actuarial Tool Design & Core ReAct Binding

Wrap your statistical calculations into functional, well-documented Python tools. Each function must have a precise, clear docstring; the LLM uses these docstrings to understand how to call your code.

- `get_sector_frequency(industry: str) -> float`: Queries the dataset and returns the empirical annual probability of a breach.

- `calculate_expected_severity(industry: str, revenue: float) -> float`: Fits a mean loss profile or record count adjusted by company scale.
- `generate_final_premium(frequency: float, severity: float, load: float) -> dict`: Computes gross premium using formal loading parameters.

Week 3: Evaluation Guardrails & Error Handling

The primary risk in agentic models is the infinite loop pattern—where an invalid input causes a code execution failure, and the agent repeatedly passes the exact same bad arguments back to the tool. You must code structural boundaries: if a tool throws an error, catch the exception cleanly and pass a helpful string back to the LLM (e.g., `"Error: Sector 'Tech' not found. Did you mean 'Technology'?"`), giving the model the precise context it needs to self-correct.

6. References & Literature

[1] Klugman, S. A., Panjer, H. H., & Willmot, G. E. (2012). *Loss Models: From Data to Decisions* (4th ed.). John Wiley & Sons. Link

[2] Werner, G., & Modlin, C. (2016). *Basic Ratemaking* (5th ed.). Casualty Actuarial Society. Link

[3] Romanosky, S. (2016). Examining the costs and causes of cyber incidents. *Journal of Cybersecurity*, 2(2), 121-135. Link

[4] Yao, S., Yu, J., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2022). ReAct: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*. Link